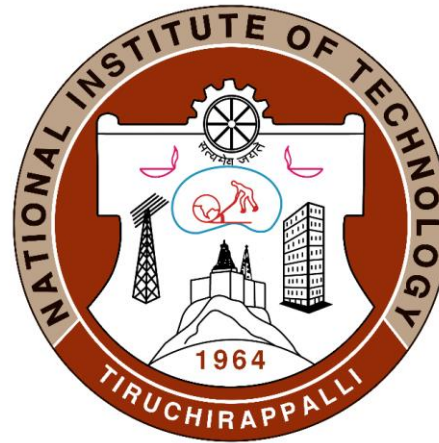


Introduction to Polymers as an Engineering Material



Dr. Nimu Chand Reger (Assistant Professor)

Metallurgical and Materials Engineering

National Institute of Technology Tiruchirappalli, Tamilnadu 620015

nimu@nitt.edu

Lecture - 10

Processing of Plastics

Contents

1. Extrusion
2. Injection Moulding
3. Blow Moulding

1. EXTRUSION

Principle

- Principle of extrusion is based on the ability of TPs to be formed into various shapes by the application of heat and pressure.
- Process of continuously pumping of plastic through a die to produce a variety of continuous forms having uniform cross-section.
- By volume 30% of TPs such as PE, PP, PVC, and PS

EXTRUSION, contd...

- Very versatile, the Forms produced: pipes, tubes, monofilaments (used for making net, rope, wire, and mosquito net), laminations, blown or cast films, strips, covering of wires and cables.
- Extrusion M/C is also used as a device for plasticizing TP material and feed other plastic forming operations such as blow moulding (which produces other hollow objects) and injection moulding.

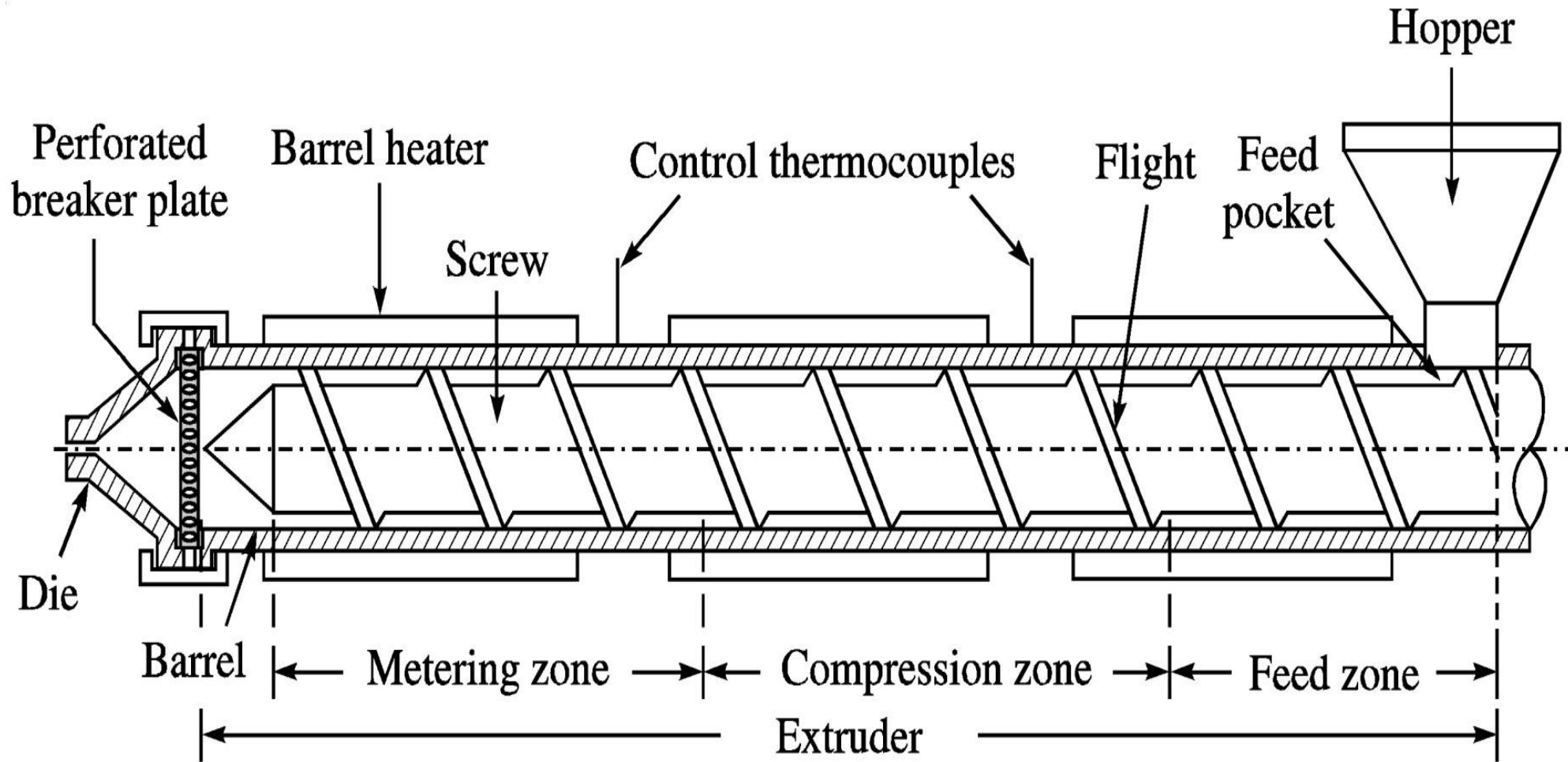


Fig. 1 Cross-section of a single-screw extrusion machine

The machine is a mixing device cum pressure vessel and basically consists of two components: an extruder and a die, as illustrated in Figure.1.

The extruder is mainly composed of a barrel with a screw rotating in it.

The Extruder barrel is further divided into 3 zones

1. Feed Zone
2. Compression zone
3. Metering (melting zone)

The Feed zone starts at the hopper end

Its function is to take the raw material (composed of granules and additives) from the hopper and transport (pump) them forward into the compression zone

In the compression Zone, as the granules move along the screw, they are plasticized by contact with a heated wall of the cylinder barrel and by the generation of heat due to friction and shear between the layers of plastics

By the time the resin enters the melting zone all the resin becomes molten
The shearing action of the screw forces the viscous melt through the die to form a continuous extrudate

At the end of the melt zone, a breaker plate having perforations is often provided to perform various useful functions

Useful functions of perforator

1. Increasing back pressure
2. Converting rotational flow of the melt into flow parallel to the screw axis
3. Holding back impurities
4. Holding back unplasticized material

After coming out from the die, the extruder is cooled immediately below its T_g to ensure the dimensional stability

2. Injection Moulding

Principle

Based on the ability of the TPs to be formed into various shapes by the application of heat and pressure.

INJECTION MOULDING, contd...

- **Used widely to form TP material.**
- **30% by vol. of TP material is produced by this Tech.**
- **Produces discrete products.**
- **Very versatile, like extrusion. Produces articles weighing as low as 25 gm to several kgs (from pin to large transport containers such as bottle crates)**
- **Intricate shaped articles can be produced by this technique.**

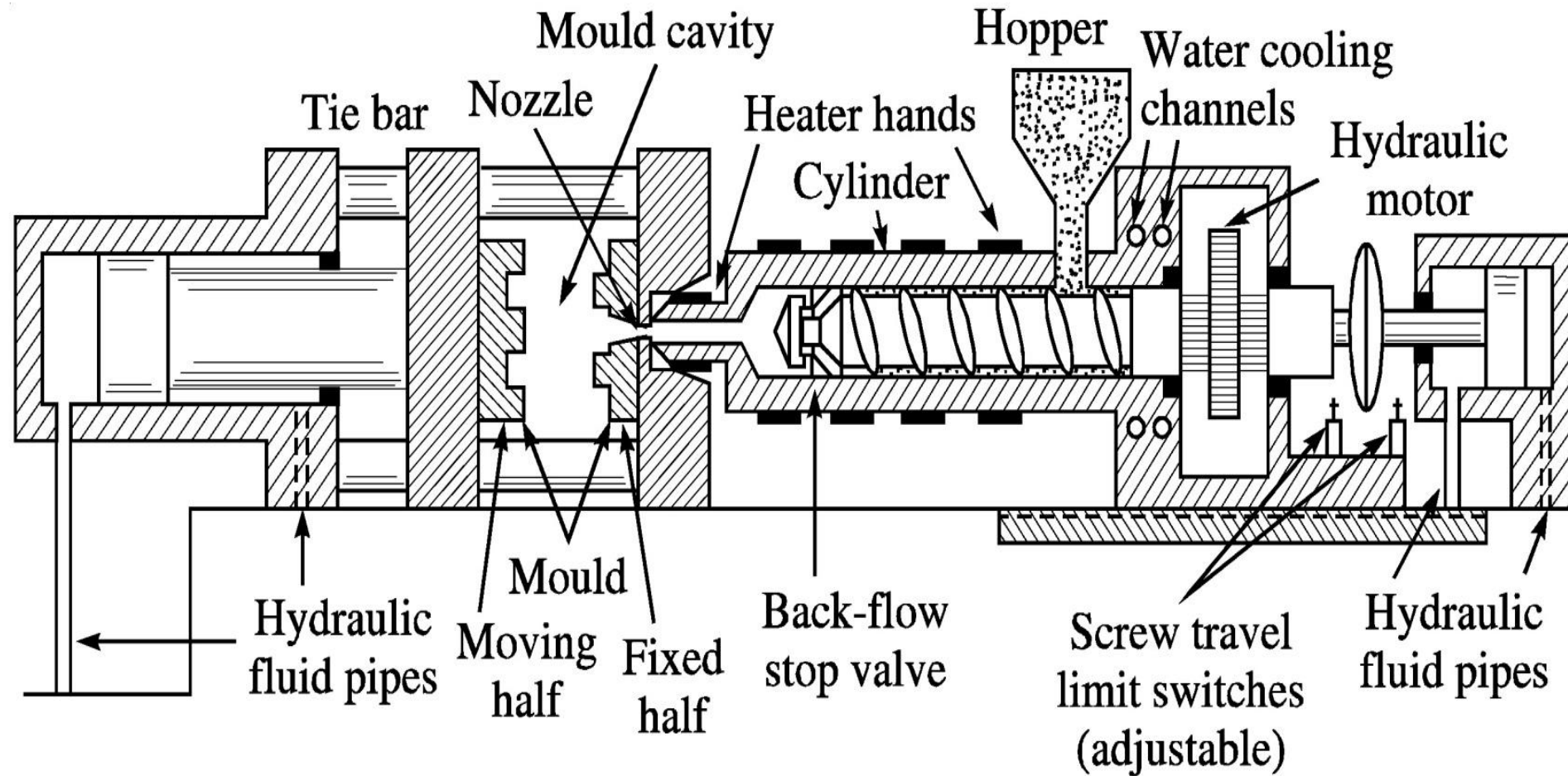


Fig. 2 Schematic diagram of reciprocating-screw injection Moulding machine

The plastic granules (or pellets) of sufficient amounts are fed through a hopper onto a rotating screw drive which rotates in a steel barrel or the cylinder as shown in Fig. 2.

As the screw moves forward it carries the charge into the heating zone of the cylinder

The rotation of the screw forces the granules against the heated walls of the cylinder

Heat of the walls and heat of compression and friction cause the plastic to soften to viscous liquid state

The screw also mixes and homogenizes the material

The temperature of the viscous melt varies from about 200-280 °C.

The higher the temperature, lower will be the viscosity of the plasticized material

Some plastics need a higher Moulding temperature while others need a lower one

The Moulding temperature also depends on the complexity of the product and the number of the pieces to be fed at a time

When enough molten plastic is collected as a pool at the mould end of the screw, the screw stops rotating and moves axially backward

Thereafter the screw acts as plunger and moves axially forward so as to inject the plasticized material through a nozzle which is in intimate contact with mould

The injected melt flows through a system of sprue, runner and gate into the closed mould cavity or cavities

The pressure is maintained on the plunger until the material in the water-cooled mould get solidified and hardened

INJECTION MOULDING, contd...

Advantages

- **A wide range of products**
- **Short cycle time**
- **Good surface finish**
- **Economic for large production runs**
- **Lower labour cost**
- **Uniform wall thickness**
- **With some modification, the M/C can be used for TSs also.**

Blow Moulding

Blow Moulding, also known as blow forming is a manufacturing process by which hollow plastic parts are formed

It is used to produce hollow components out of thermoplastic



BLOW MOULDING

Principle

- Expansion of a tube of semi molten plastic material called parison using compressed air until it conforms to shape of the mould around it.
- Used to produce hollow products from practically any thermoplastic material, viz. LDPE, HDPE, PS, PS, PVC, P, PET, etc.

Blow molded parts can be formed from a variety of thermoplastic

1. Low density polyethylene (LDPE)
2. High density polyethylene (HDPE)
3. Polyethylene Terephthalate (PET)
4. Polypropylene (PP)
5. Polyvinyl Chloride (PVC)

Advantages

1. **Moulding extremely irregular (reentrant) curves**
2. **Low stresses in the product**
3. **Possibility of variable wall thickness**
4. **Use of polymers with high chemical resistance**
5. **Favorable processing costs**

The overall process comprises of four steps

1. The production of a preform or the so-called parison
2. Bringing the parison within the mould
3. Forming the neck of the component depending on the shape of container
4. Blowing the parison to the shape of the mould cavity and simultaneous cooling of the mould and Moulding

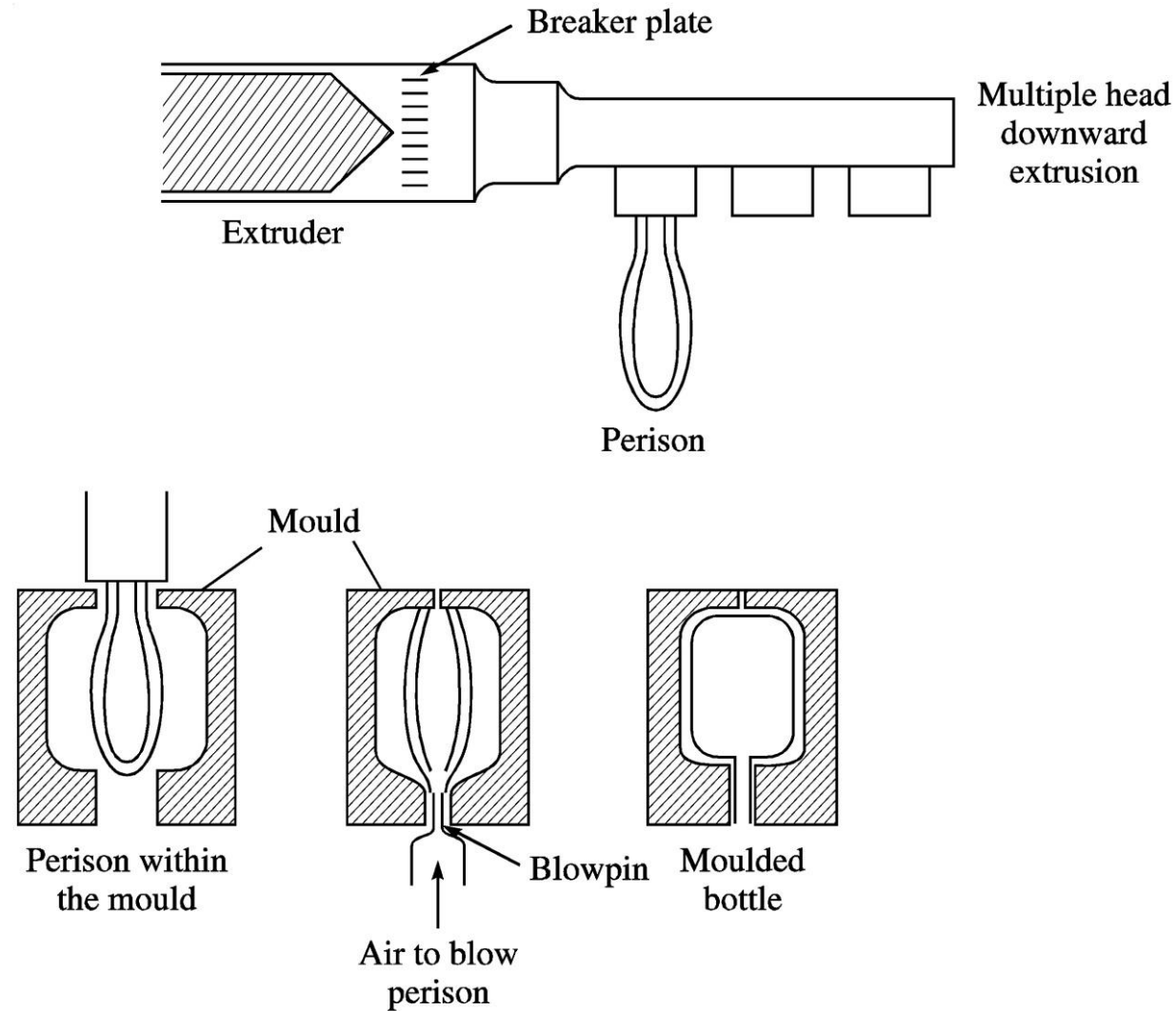


Fig. 3 Stage of extrusion blow molding of plastic bottles

TYPES OF BLOW MOULDING

1. Extrusion Blow Moulding (EBM) which principally uses an unsupported parison (75% usages)
2. Injection Blow Moulding (IBM) which principally uses a preform supported by a metal core pin (25% usages).
3. Stretch Blow Moulding either EBM or IBM to obtain bioriented products.

Extrusion Blow Moulding

Machinery

1. An Extruder
2. A die head
3. A Blow Mould

• Process

- ☐ Melting the Plastics in an Extruder.
- ☐ Forming the parison (A hot plastics tube).
- ☐ Clamping the two mould halves around the parison and keeping closed.
- ☐ Inflating the parison inside the Blow (female) mould.
- ☐ Cooling of the expanded parison
- ☐ Opening

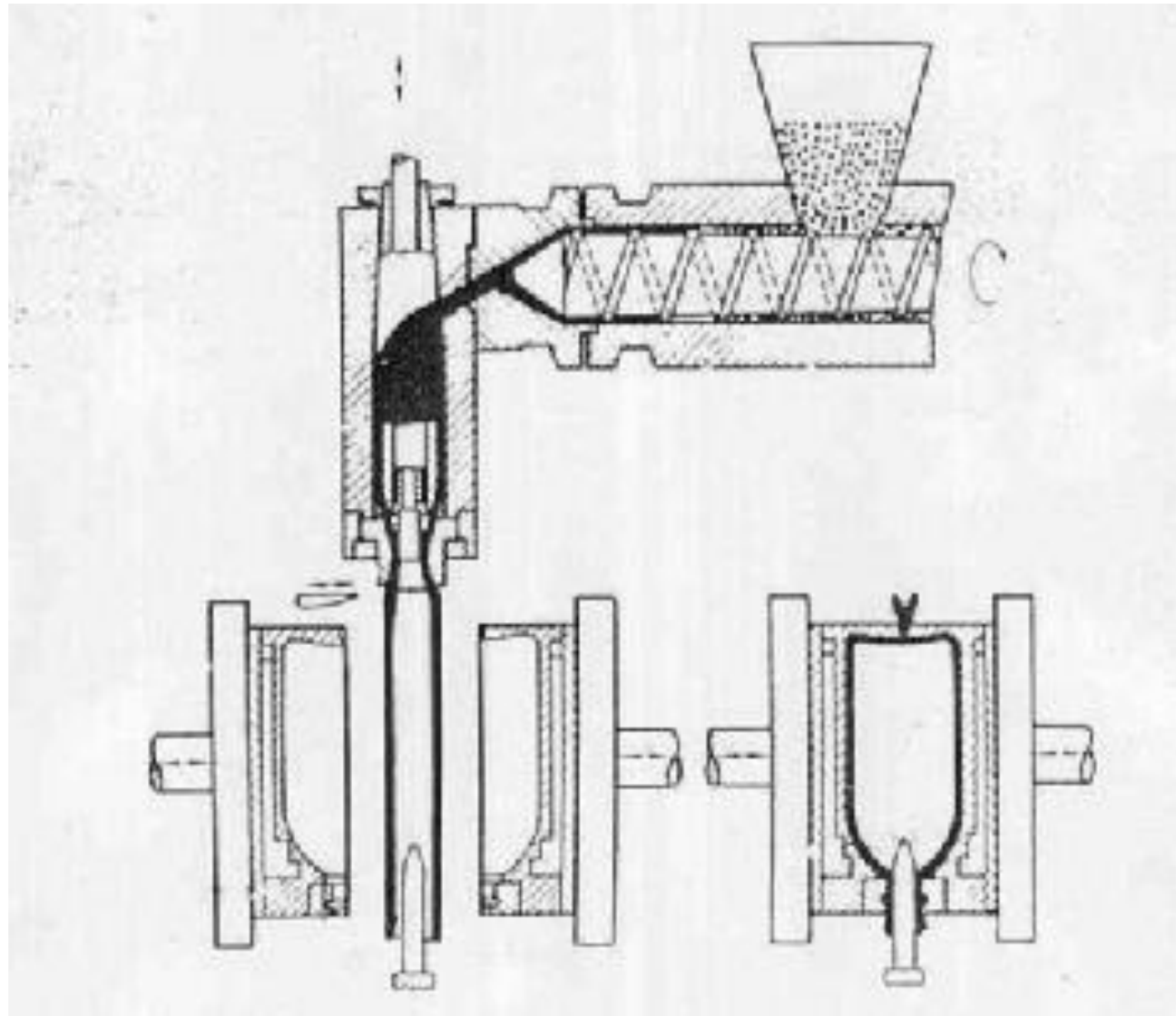


Fig. 4 Typical blow moulding set up

Advantages

- High rate of production.
- Low tooling costs.
- Incorporation of blown handle ware.
- A wide selection of machine manufacturers.

Disadvantages

- ❖ Higher Scrap Rate.
- ❖ Use of recycled scrap.
- ❖ Limited wall thickness control.

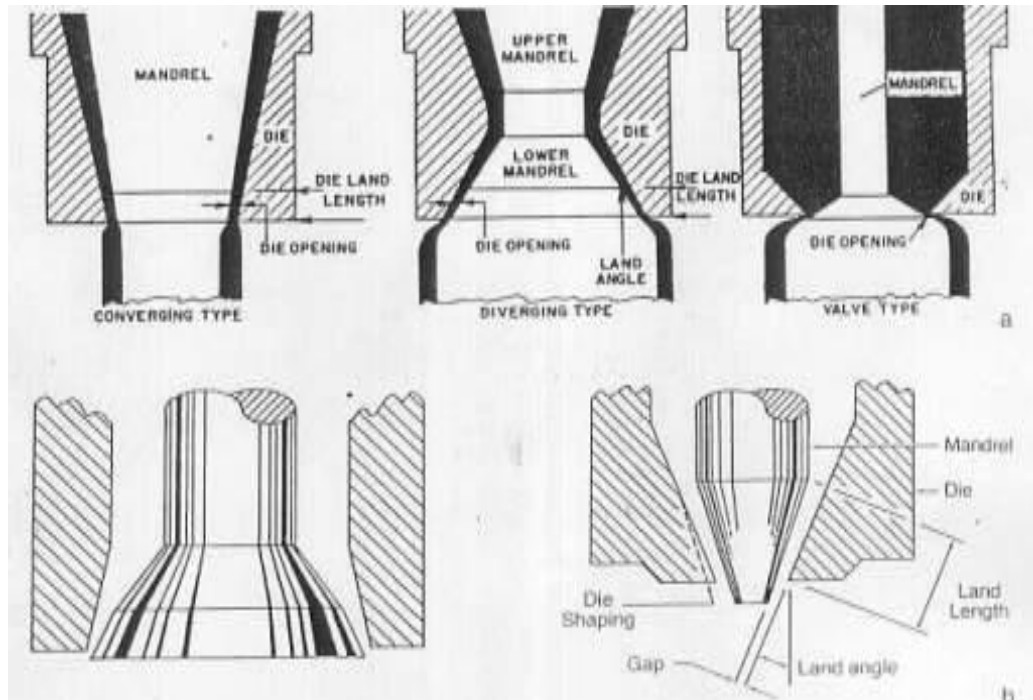


Fig. 5 Basic type dies to form parison

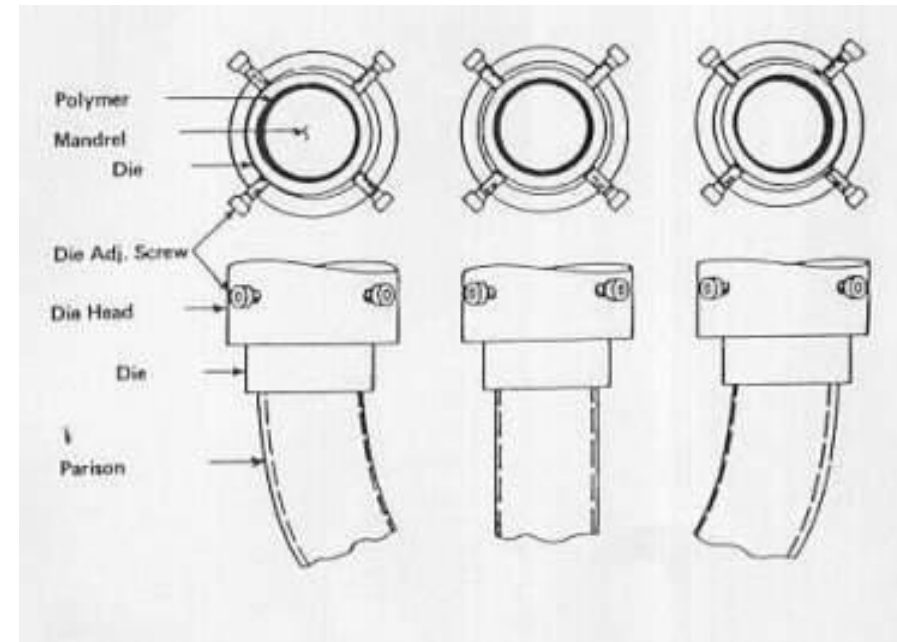


Fig. 6 Adjusting die tools to obtain proper parison shape

INJECTION BLOW MOULDING

Injecting a preform and Blowing the same.

Process Steps

- 1. Hot melt in exact amounts is injected through an injection moulding machine nozzle into one or more preform cavities.**
- 2. Hot liquid is circulated through mould heating channels around the preform cavity and the required preform temperature is achieved.**
- 3. The two-part mold opens, and the core pin carries the hot plastic to the second stage blow mold station.**
- 4. Upon mold closing, air is introduced via the core pin and the plastic blows out and contacts the mold cavity surface.**
- 5. The product cools and the blow mould opens, and the product is ejected.**

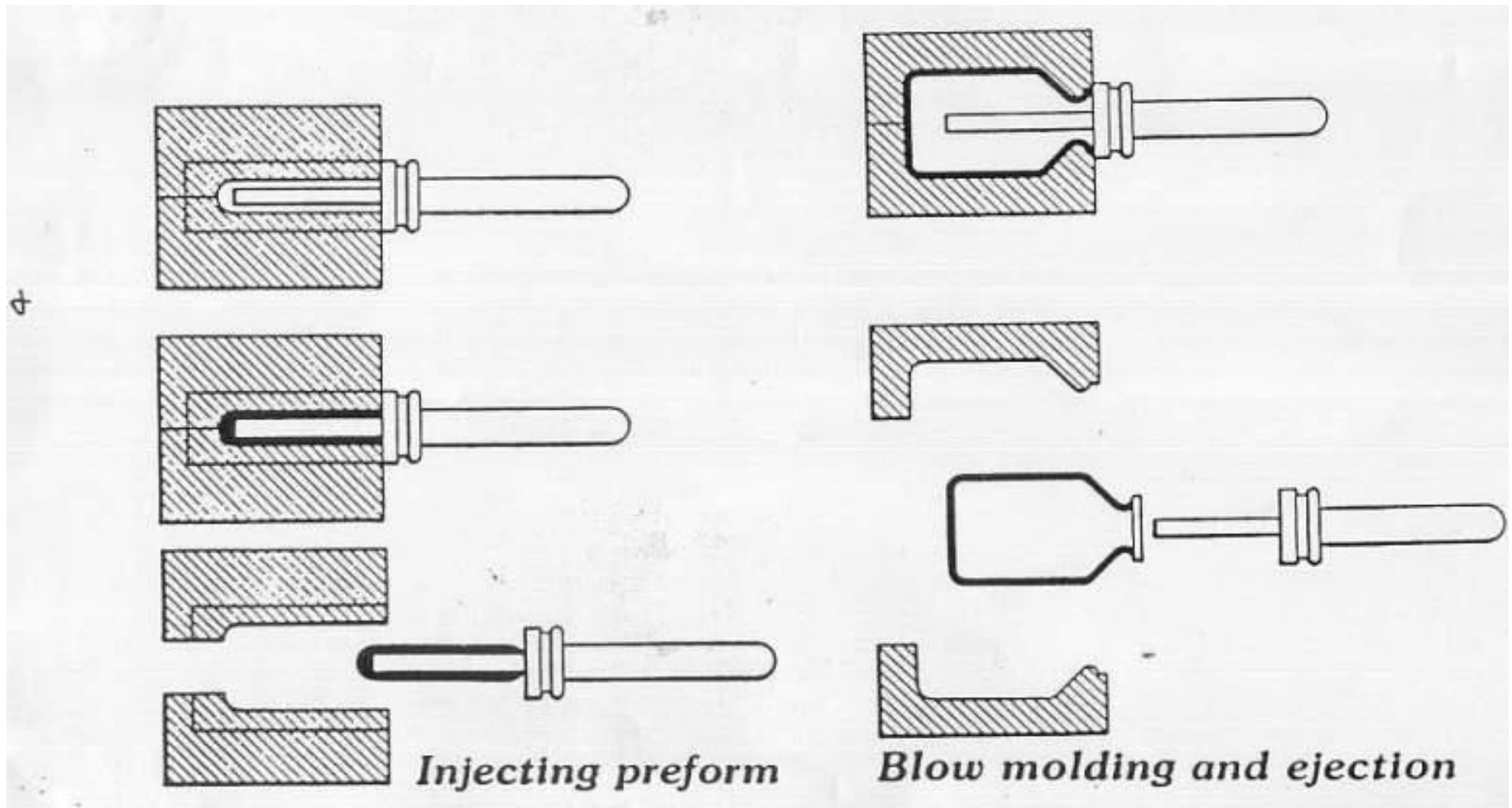


Fig. 7 Basic Injection blow molding process

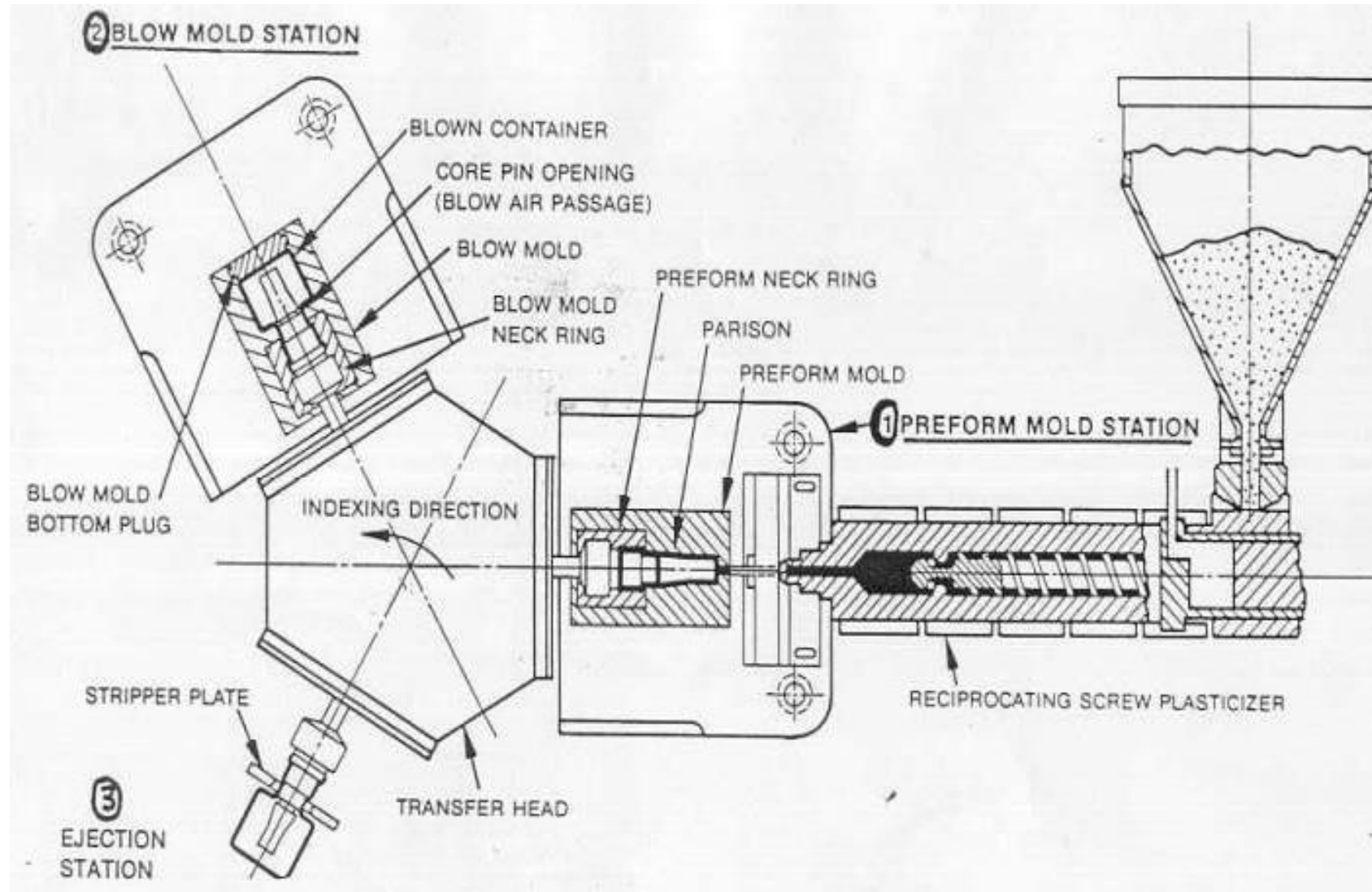


Fig. 8 Schematic of a three-station injection blow molder

Advantages

1. Bottle necks of varying shapes, sizes and wall thickness can be precisely molded, Free of flash.
2. The wall thicknesses of a hollow body can be predetermined exactly.
3. There is no seam on the bottom.
4. There is no regrind to contend with, other than that generated on state-up and rejected parts, because there is no neck on bottom trim.
5. It can be used to blow mold almost any type of plastics that can be injection molded.

Disadvantages

1. High change-over costs from one part to another. This is caused by the large number of molds and core pins required.
2. Size limitations on parts which can be made by blow molding.
3. Tooling costs are almost double that of extrusion blow molding.

STRETCH BLOW MOULDING

- * Biaxial Orientation Blow Moulding.**
- * The container is stretched radially by the blow air and vertically by a mechanical stretch rod.**

Advantages

- Better Strength**
- Better Clarity**
- Increased Impact Strength**
- Improved gas and water vapour barrier**
- Reduced Creep**

Materials used

PET

PAN

PVC

PP

ABS

ACETALS

PS

Machine Categories

- (i) In-line machines
- (ii) Two-stage machines

Inline machines

In this process an extruded or injection molded parison passes through conditioning stations which bring it to the proper orientation temperature and then stretched and blown in a final stage blow mould.

Advantages

1. Heat history is minimized (important for heat-sensitive materials like PVC and PET)
2. The preform can be programmed for optimum material distribution when maintained under continuous control.

Two-stage machines

In this process the extruded or injection moulded preforms that have been cooled are conveyed through an oven which reheats them to the proper orientation-blow temperature.

Advantages

1. Minimized scrap production
2. Improved thread finish
3. Higher output rates
4. The capability to stockpile preforms, eliminating dependency between preform and bottle machinery.

TYPES OF STRETCH BLOW MOULDING

- * Injection Stretch Blow Moulding.**
- * Extrusion Stretch Blow Moulding.**

EXTRUSION STRETCH BLOW MOULDING

- Suitable for processing of PVC, PET.**

Process

- § Extruding a parison.**
- § Converting the parison into a preform.**
- § Blowing and stretching the preform in the final blow mould.**

EXTRUSION STRETCH BLOW MOULDING

Suitable for processing of PVC, PET.

Process

- Extruding a parison
- Converting the parison into a preform.
- Blowing and stretching the preform in the final blow mould.

EXTRUSION STRETCH BLOW MOULDING

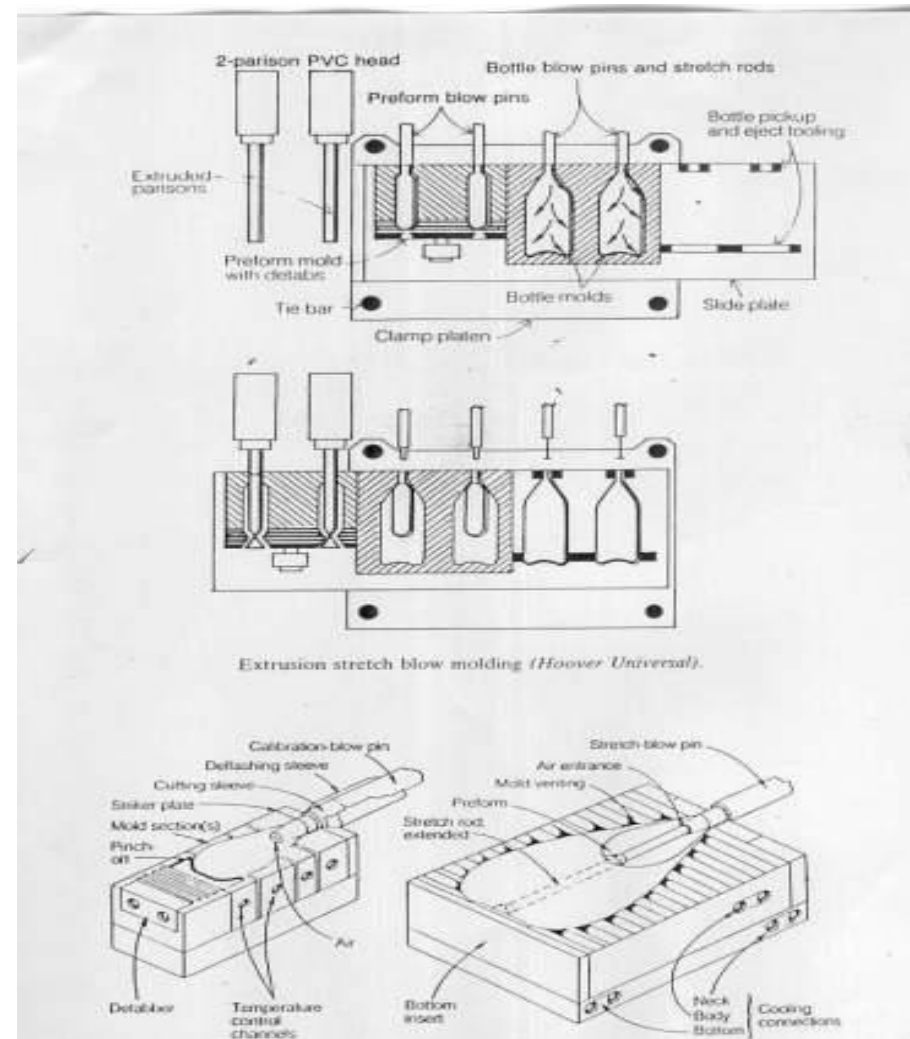


Fig. 9 Various segments of extrusion stretch blow moulding

INJECTION STRETCH BLOW MOULDING

Suitable for processing of PET

Process

- 1. Injecting a preform.**
- 2. Reheating the preform if necessary.**
- 3. Blowing and stretching of the preform in the final blow mould.**

INJECTION STRETCH BLOW MOULDING

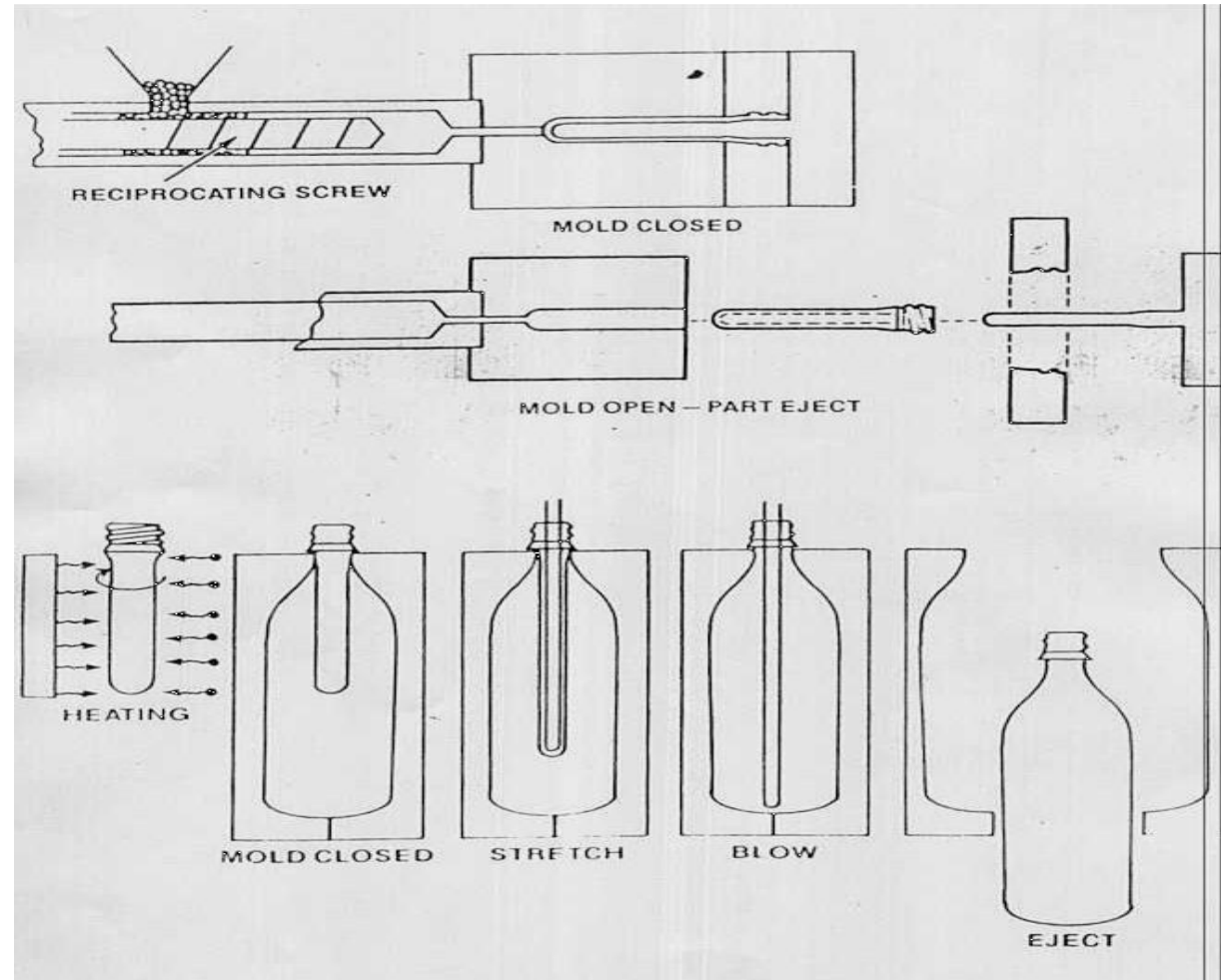


Fig. 10 Schematic view of injection stretch blow moulding

Stretch blow moulding processing

Type of plastic	Melt temp		Stretch Orientation Temp		Maximum Stretch ratio
	°F	°C	°F	°C	
PET	490	250	190-240	88-116	16/1
PVC	390	199	210-240	99-116	7/1
PAN	410	210	220-260	104-127	9/1
PP	334	168	250-280	121-136	6/1

A temperature conditioned injection preform is inserted into the blow mould cavity, then is rapidly stretched. Often a rod is used to stretch the preform in the axial direction with air pressure to stretch the preform in the radial direction

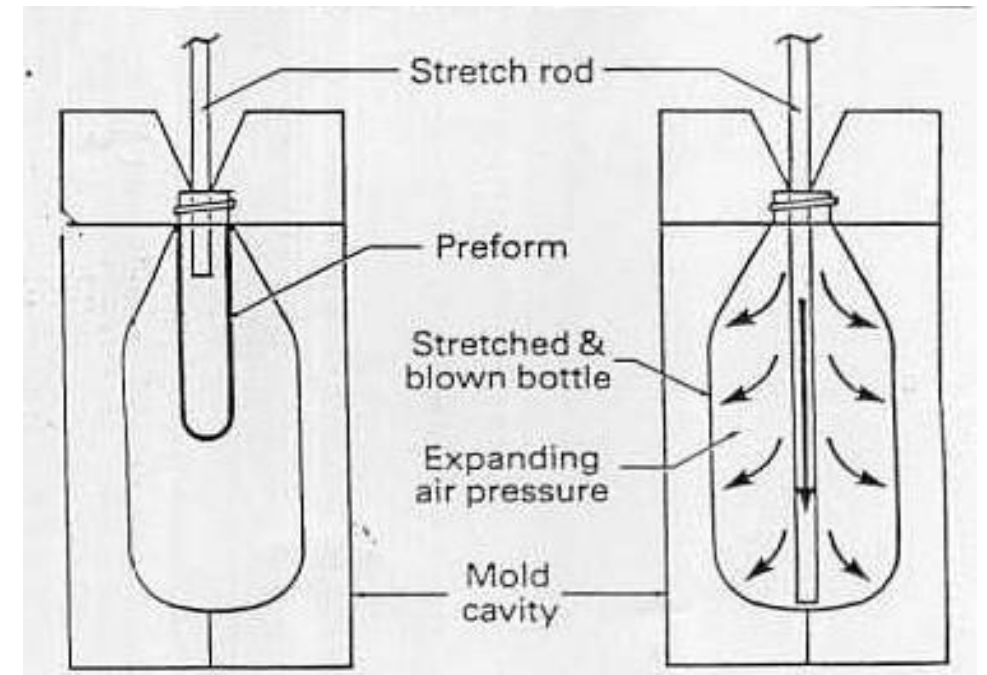


Fig. 11 Schematic view of stretch blow moulding

Factors to consider when reviewing the complete blow moulding process

1. Part design

Part configuration (size/shape)-relate shape to flow of melt in mould to meet performance requirements, which should at least include tolerances

2. Material

Chemical structure; molecular weight amount and type of fillers/additives; heat history; storage; handling

3. Mould design

Number of cavities; layout and size of cavities cooling lines/side actions/knockout rins/etc-relate layout to maximize proper performance of melt and cooling flow patterns to meet part performance requirements. Pre-engineer design to minimize wear and deformation of mould (use proper steels, Al, etc) and lay out cooling lines to meet. Temperature to time cooling rate of plastic (particular crystalline types)

4. Machine Capabilities

Accuracy and repeatability of temp/time/velocity pressure controls of extrusion injection unit, accuracy and repeatability of clamping force. Flatness and parallelism of platens; even distribution of clamping : repeatability of controlling pressure and temperature of oil; minimize oil temperature variation.

5. Moulding Cycle

Setting up the complete moulding cycle to repeatedly meet performance at the lowest cost by interrelating material/machine/mould controls such as those listed in this chapter

Guide to processing temperatures of plastics for blow moulding

Plastic	Temperature °C
LDPE	130-180
MDPE	150-200
HDPE	160-220
HMWPE	180-230
PVC	190-205
PP	200-220
PS	280-300
PA	240-270
POM	150-280
SB	170-210
ASA	200-230

Guide for Air Blowing Pressure

Acetal	100-150
PMMA	50-80
PC	70-150
LDPE	20-60
HDPE	60-100
PP	75-100
PS	40-100
PVC (rigid)	75-100
ABS	50-150



THANK YOU